

Adaptive Traffic Light Control Using a Distributed Processing Approach

Anton Agafonov
Samara National Research University
Samara, Russia
ant.agafonov@gmail.com

Abstract— This paper deals with a problem of an adaptive control of traffic lights in an environment of connected vehicles. An algorithm founded on maximizing of the predicted weighted flow of traffic through the intersection at a given phase of the traffic light was the investigated matter. The algorithm implementation aimed at high-performance traffic processing systems. To take advantage of both stream and batch-processing of traffic data, the lambda data-processing architecture and Apache Spark framework were used in the construction of the algorithm. An experimental study of the adaptive traffic light control algorithm is conducted on the traffic data from the SUMO microscopic traffic simulation software. Conclusions are drawn that the developed algorithm can be effectively used for high-performance traffic processing in a large-scale traffic scenario.

Keywords—adaptive control, traffic light control, intelligent transport systems, big data, Spark, SUMO

I. INTRODUCTION

An adaptive traffic flow control problem is a part of the key problems improved the efficiency of using the transport infrastructure in transport systems. There are various methods to control traffic flows that can be implemented as a part of an intelligent transportation system: anticipatory vehicle routing [1], restricting access into congested areas, adaptive traffic signal control (ATSC) [2], etc. This article considers the ATSC problem. Adaptive control is the selection of traffic light phases according to the current (observed) situation on roads.

The ATSC problem is an actively developed area of research. Reinforcement learning methods and algorithms to solve a control problem are available today owing to the development of machine learning (ML) methods. The main sources of vehicle traffic data are video surveillance systems at intersections and motion sensors on traffic lanes. In the future, the development of autonomous and connected vehicles and the implementation of Vehicular Ad-hoc Networks (VANETs) will enable the use of large amount of traffic data to improve the efficiency of traffic management. Meanwhile, the existing solutions to the control problem (both traditional and based on ML methods) do not consider large-scale traffic scenarios, the issues of processing large amounts of data and controlling traffic signals in large urban areas.

In compliance with main goal of this paper, it is important to study and implement the ATSC method that consists in maximizing the traffic flow [2] as part of a high-performance intelligent traffic processing system. The developed solution should be horizontally scalable so as to be capable of controlling traffic lights in a large-scale vehicle traffic scenario.

The paper is structured as follows. The literature on the subject of the article is given in Section II. Section III describes the method of adaptive traffic light control. Section IV outlines a system architecture using the MapReduce approach to high-performance processing of traffic data. Section V presents the results of experimental studies of the implemented algorithm. Conclusion and proposed directions for further research complete the paper.

II. LITERATURE REVIEW

In [3], the authors presented an overview of classical and ML-based methods for management of traffic light, including coordination strategies. An overview of ATSC methods in an environment with connected and autonomous vehicles is presented in [4]. The authors compared existing control methods, described the advantages and disadvantages of the approaches, and showed that the use of data from connected vehicles can reduce traffic congestion, travel time, and waiting time at intersections.

In [5], a deterministic approach was used based on maximizing the "pressure" at the intersection, i.e. the distance between vehicles in inbound and outbound lanes at a junction. In [1], the authors proposed a control method based on maximizing the predicted weighted flow of vehicles moving across an intersection. The phase of the traffic signal cycle was chosen for which the predicted flow is maximum. In [6], the authors performed short-term traffic flow forecasting for each intersection. The minimization of blocked vehicles was used as an optimization function in the traffic light control problem. In [7], the authors used data from probe vehicles to implement coordinated control of traffic signals on arterial roads. In [8], data from IoT devices was used to extract features, classify, and control traffic at intersections as a part of the Smart City. In [9], the authors used a fuzzy logic approach to optimize the traffic signal cycle calculated using the Webster formula.

Currently, much attention from researchers has attracted the use of ML methods and, especially, reinforcement learning (RL), to solve the ATSC objective. In [10], the authors described the fundamental concepts of RL algorithms and its application to adaptive control tasks, considered the autonomous and connected vehicles effect on control of traffic light, and discussed open questions and future research directions. In [11], RL methods and algorithms for the traffic control problem were considered, including the representation of RL models (state, action, and reward), comparison metrics, and unsolved problems.

In practice, it is possible to divide the RL-based algorithms into two main groups of methods according to its foundation: value and policy. Value-based RL methods explore an action utility function (Q-function). In addition, the researchers used a deep Q-learning approach to assess the effect of phase duration in [12]. The proposed approach minimized both the waiting time when the traffic light is red and the losses while

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the vehicle braking and accelerating. Using Deep Q-Network, the researchers in [13] select the traffic light phase and its duration simultaneously. In [14], a double Q-learning approach was used; a traffic flow forecast was used as one of the state space parameters.

Policy-based RL methods are aimed at directly studying the function of a policy, which determines the likelihood of performing a certain action in the conditions of a specific state [15-17]. These separate approaches can be combined into a method called Actor-Critic, where Actor scrutinizes the policy that determines what action the agent performs, and Critic evaluates how successful this action is using a utility function [18-20].

It should be noted that in most works, the analysis of the developed models is carried out at one intersection or in a small area, and the delays in computations are not taken into account. Moreover, RL methods can be unstable under certain conditions and computationally-demanding for training. The goal of this study is to investigate the ATSC algorithm implemented as part of a high-performance traffic control system.

III. METHOD OF ADAPTIVE CONTROL OF TRAFFIC LIGHTS

To solve the traffic light control problem in a context of a high-performance intelligent traffic processing system, the deterministic method of maximum weighted flow (MaxPWFlow) [1] proposed by the author of this paper is applied. The traffic light next phase is chosen based on the predicted flow of vehicles passing through the intersection for the considered time interval. It is selected that phase, which corresponding to the maximum predicted flow.

Let us present a brief description of the used method. Introduce the following notation. Let P be a set of phases of the considered traffic light, $phase$ be the next selected phase, τ_{min} be a phase switching interval, t_p , $p \in P$ be a duration of the current active phase of the traffic light.

Using this notation, the MaxPWFlow algorithm can be described in a pseudocode form as follows (Algorithm 1).

Algorithm 1: MaxPWFlow method

Input data: τ_{min} , t_p , p , P

Output data: $phase$

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if  $t_p < \tau_{min}$  then
     $phase = p$ 
     $t_p = t_p + 1$ 
else
     $phase = \text{argmax}(\{PWFlow(p) \text{ for } p \text{ in } P\})$ 
     $t_p = 0$ 
end if

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The main operation of the method is the estimation of the predicted weighted traffic flow (i.e. number of vehicles) running through the intersection for a given phase $p \in P$. The traffic flow estimation is performed using the function $PWFlow(p)$.

Introduce the additional notations. Let L_{phase}^{income} be the multiplicity of lanes that are allowed in connection with

the phase $phase$ that corresponds to the green light of the traffic light, C_l be a set of vehicles moving in a lane $l \in L_{phase}^{income}$. Then the $PWFlow(p)$ function can be defined as follows:

$$PWFlow(phase) = \sum_{l \in L_{phase}^{income}} \sum_{c \in C_l} \eta(c, l) I(t(c) < \tau_{min}), \quad (1)$$

where $t(c)$ represents a time estimate required by the vehicle $c \in C_l$ to reach the intersection, $I(val)$ is an indicator, returning 1 if $val = True$, and 0 otherwise.

A deterministic predictive model [1] is applied to calculate $t(c)$ value that is needed for vehicle $c \in C_l$ to pass the intersection.

The coefficient $\eta(c, l)$ needed to reckon in the waiting time $wait(c, l)$ of the vehicle c in the lane l as follows:

$$\eta(c, l) = 1 + \alpha \cdot wait(c, l), \quad (2)$$

where $\alpha = 0.01$ is an empirically chosen coefficient.

It should be noted that all calculations are performed at the moment of phase switching and are usually considered instantaneous. Of course, in practice, such assumptions cannot be used. As a result, implementation the developed algorithm as a part of a high-performance traffic processing system as described in the next section was proposed.

IV. PROPOSED APPROACH

To implement the developed algorithm, a lambda architecture was chosen. It is a data-processing architecture that consists of three layers:

- Stream-processing layer that performs real-time data processing.
- Batch-processing layer that performs computationally consuming calculations.
- Serving layer that responding to queries for the next traffic light phase.

The proposed data processing pipeline consists of the following steps:

- Each vehicle sends information about its movement (position, speed, acceleration, and route). This data is fed into Apache Kafka, which is an open source distributed software broker of streaming messages.
- At the stream-processing layer (every second), vehicle data is processed and stored in RAM memory.
- At the batch processing layer, travel time estimation for each vehicle is performed and the next phase for each traffic light that maximizes the predicted traffic flows is chosen (every two seconds). The results are stored in a database.
- The next phases are requested and returned from the database at the serving layer.

To implement the proposed algorithm using the described architecture, it is applied the Apache Spark framework that is widely used for large-scale data processing. As the database for processing result storage, the NoSQL database Apache Cassandra was chosen. The architecture of the designed system is presented in Fig. 1.

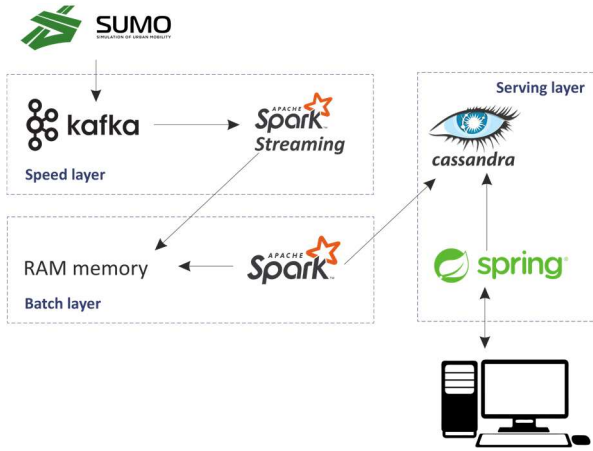


Fig. 1. Architecture of the designed system.

All components of the proposed system support horizontal scaling, i.e. it is allowed to add additional computational nodes to the cluster in case of necessity have to handle scenarios with larger traffic.

Furthermore, it should be mentioned that the calculation of the traffic light next phase and the return of phases on requests are performed independently. As a result, there are no additional delays to perform the control of traffic light.

V. EXPERIMENTAL STUDIES

The SUMO (Simulation of Urban Mobility) microscopic traffic simulation software [21] is used to determine the effectiveness of the distributed implementation of the MaxPWFlow method (denote below as MaxPWFlow-DI). This open-source package is designed to simulate vehicle movement in large-scale transportation networks.

The experimental study was carried out on a large-scale road scenario, where the number of intersections of various configurations was 2928, which includes 316 traffic light-controlled intersections. The modeling of traffic was executed in period from 7 am to 9 am, when a large congestions were observed.

The road network of the considered scenario is shown in Fig. 2.

As baseline algorithms for comparison, the following classical and RL-based methods and algorithms were used:

- Independent Proximal Policy Optimization (IPPO) [15]
- Independent DQN (IDQN) [16], a value-based deep Q-learning RL-method.
- Advantage Actor-Critic (A2C) [18].
- MaxPressure [5].
- MaxPWFlow [1] – non-distributed implementation of the developed algorithm.

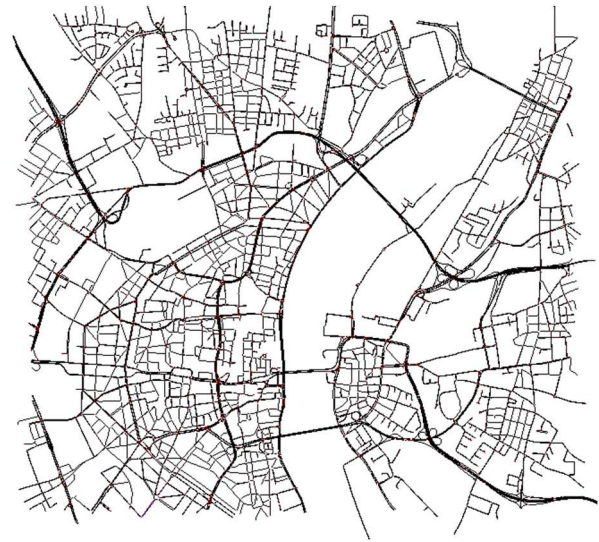


Fig. 2. Road network under consideration.

To define quality of the considered traffic light control algorithm, following criteria were used:

- The time required for vehicles to end their route defined as the average travel time measured in seconds.
- The time a vehicle was standing at an intersection is defined as the average waiting time measured in seconds.

To calculate the described criteria values, the simulation was run 10 times (episodes) and the results were averaged over the episodes. Each episode has different initial vehicle distribution (initial positions, departure times, and routes). The results of the comparison according to the specified criteria are given in Table I.

TABLE I. COMPARISON OF ALGORITHMS

	Travel Time, s	Waiting Time, s
IPPO	348.8	32.8
IDQN	329.9	15.1
A2C	348.2	26.5
MaxPressure	337.9	20.3
MaxPWFlow	328.5	17.4
MaxPWFlow-DI	326.6	21.9

Upon the research results, it can be concluded that the distributed implementation of the MaxPWFlow method outperforms other baseline algorithms by the "travel time" criteria but underperform by the "waiting time" criteria.

It should be noted that the proposed architecture supports high-performance traffic processing, independent processing for multiple traffic lights, and horizontal scaling.

VI. CONCLUSION

In this paper, a lambda data processing architecture has been designed to implement the method for adaptive traffic light control called Max Predicted Weighted Flow in a distributed environment. This method is aimed to maximize the predicted traffic flow via the intersection at a specific phase of traffic light.

Experimental study of the proposed implementation evaluated using the SUMO simulation software. The comparison of the classical and reinforcement learning-based

methods is conducted in a real-world traffic scenario. According to the considered criteria of the average waiting time and the average travel time, the algorithm showed the best results for the first criterion, while it was inferior to other methods for the second criterion.

The next stage of work will include the implementation of more complex travel time prediction algorithms and, in particular, short-term traffic forecasting using deep neural networks.

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